

Lecture 2: Kirchhoff's Laws

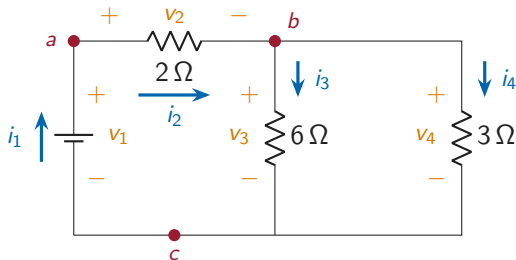
EE201 – Circuit Theory I

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■ current ■ voltage ■ power ■ topology

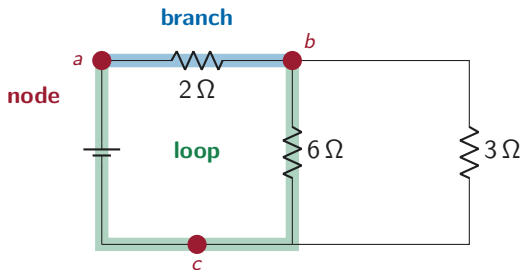
A circuit — and what we do not yet know



Four branches, each carrying a voltage and a current:

$$v_1, v_2, v_3, v_4 \quad \text{and} \quad i_1, i_2, i_3, i_4 \quad \Rightarrow \quad 2b = 8 \text{ unknowns}$$

Node, branch, loop



Node — a connection point; everything joined by ideal wire is *one* node, no matter how much wire is drawn. Here $n = 3$.

Branch — one element, two nodes. Here $b = 4$.

Loop — a closed path through no node twice.

What each element says — on its own



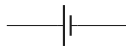
$$v = R i$$



$$v = L \frac{di}{dt}$$



$$i = C \frac{dv}{dt}$$



$$v = v_s(t)$$

For our circuit: $v_1 = 12 \text{ V}$, $v_2 = 2 i_2$, $v_3 = 6 i_3$, $v_4 = 3 i_4$

That is 4 equations for 8 unknowns. Exactly half —
and staring harder at the elements will never produce the rest.

Two kinds of equations — and they never mix

Terminal equations

$$v = Ri, \quad i = C dv/dt, \dots$$

b of them

*what each element is
— it does not know where it is*

+

Network equations

KCL and KVL

b of them

*how they are wired
— it does not know what they are*

Because KCL and KVL ignore the elements, they hold for **every** lumped circuit — linear or not, time-invariant or not, at DC or at 100 MHz.

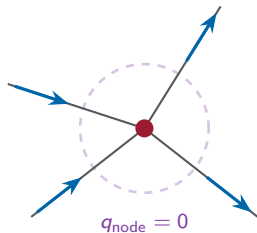
Kirchhoff's Current Law

$$\sum_{k \in \text{node}} \pm i_k(t) = 0$$

at every node, at every instant.

What flows in, flows out.

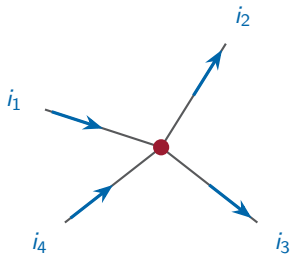
Why: a node cannot store charge



$$\frac{dq_{\text{node}}}{dt} = \sum_{\text{in}} i_k - \sum_{\text{out}} i_k = 0$$

Conservation of charge, plus *nothing piles up* — the first lumped assumption from Lecture 1.

Sign rule, and an example

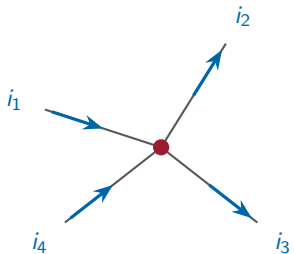


Leaving positive:

$$-i_1 + i_2 + i_3 - i_4 = 0$$

Given $i_1 = 3 \text{ A}$, $i_2 = -2 \text{ A}$, $i_3 = 6 \text{ A}$, what is i_4 ?

Sign rule, and an example



Leaving positive:

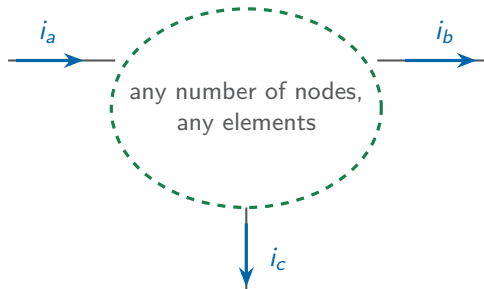
$$-i_1 + i_2 + i_3 - i_4 = 0$$

Given $i_1 = 3 \text{ A}$, $i_2 = -2 \text{ A}$, $i_3 = 6 \text{ A}$, what is i_4 ?

$$-3 + (-2) + 6 - i_4 = 0 \implies i_4 = 1 \text{ A}$$

A measured -2 A goes in *with its sign*. We never redraw the arrow.

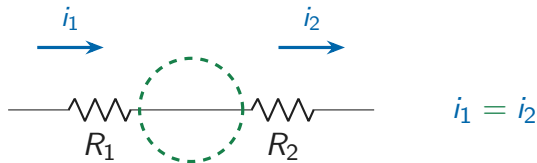
Nothing in the argument needed a point



$$-i_a + i_b + i_c = 0$$

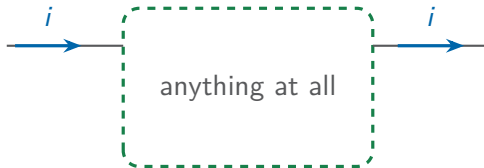
Only the branches that **cross** the surface appear. One node or a thousand, resistors or a whole amplifier — it makes no difference.

Consequence I: series means one current



Exactly two branches cross the surface. A chain of elements has **one** current — and that *is* the definition of “in series”.

Consequence II: a box gives back what it takes



This is what lets us say “**the** current through” a black box — and what makes a two-terminal element a meaningful idea at all.

Two things KCL is not

Not averaged

$\sum \pm i_k(t) = 0$ holds at each instant t *separately*. Nothing is averaged over time.

Not about v

It says nothing about voltage, resistance or power. Topology and current only.

Kirchhoff's Voltage Law

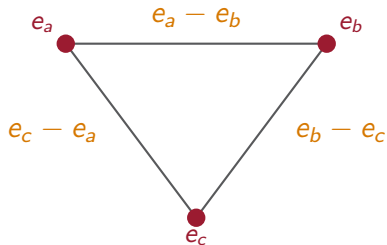
$$\sum_{k \in \text{loop}} \pm v_k(t) = 0$$

round **every loop**, at **every instant**.

Go all the way round and you come back to where you started.

As with KCL, this is the lumped model at work: it holds as long as no magnetic field is changing through the loop — assumption (b) from Lecture 1.

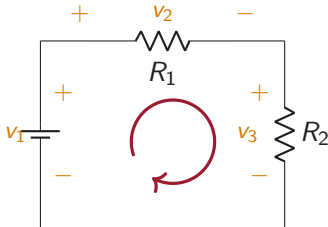
Why: the differences telescope



$$(e_a - e_b) + (e_b - e_c) + (e_c - e_a) = 0$$

Every node carries one number. A branch voltage is a *difference*. Around a loop they cancel in pairs.

Sign rule: walk the loop

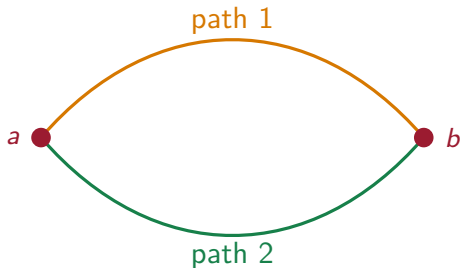


$$-v_1 + v_2 + v_3 = 0$$

- + when you traverse a branch from + to -;
- - when you go the other way.

Walking anticlockwise multiplies every term by -1 — same equation.

The same statement: voltage forgets the path



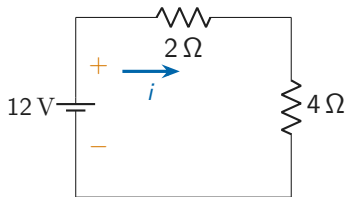
$$V_{ab} = V'_{ab}$$

Path 1 forwards then path 2 backwards is a loop, so the two must agree.

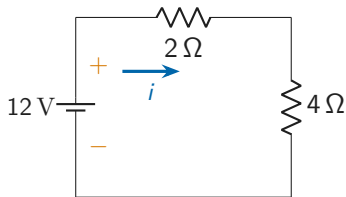
This is why “the voltage across R_2 ” means something — and why node potentials exist at all.

Example: one loop

Walk the loop. What is i ?



Example: one loop

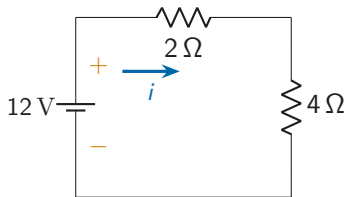


Walk the loop. What is i ?

$$-12 + 2i + 4i = 0$$

$$\Rightarrow i = 2\text{ A}$$

Example: one loop



Walk the loop. What is i ?

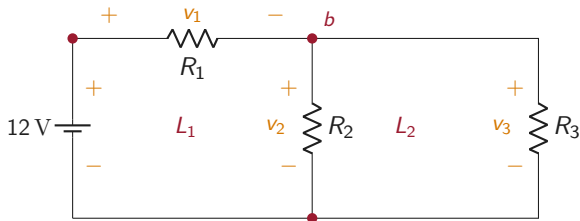
$$-12 + 2i + 4i = 0$$

$$\Rightarrow i = 2\text{ A}$$

$$v_{2\Omega} = 4\text{ V}, \quad v_{4\Omega} = 8\text{ V}$$

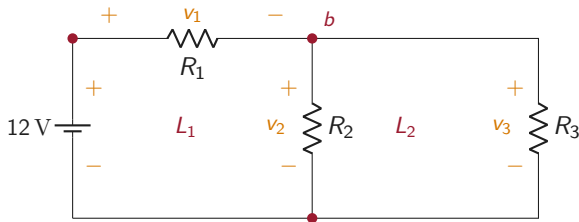
$4 + 8 = 12$: the source's volts are shared in proportion to resistance.

Example: three loops, two equations



$$L_1 : -12 + v_1 + v_2 = 0 \quad L_2 : -v_2 + v_3 = 0$$

Example: three loops, two equations

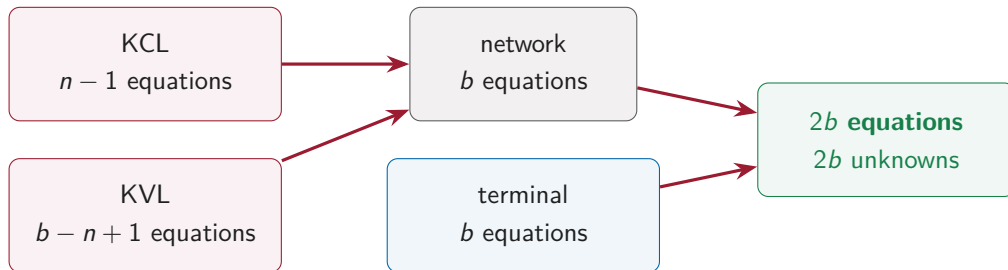


$$L_1 : -12 + v_1 + v_2 = 0 \quad L_2 : -v_2 + v_3 = 0 \quad \underbrace{L_3 : -12 + v_1 + v_3 = 0}_{= L_1 + L_2}$$

The third loop is **true but not new**. Only $\ell = b - n + 1 = 2$ of them count.

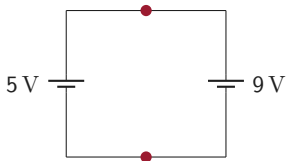
The whole bookkeeping, on one slide

n = number of **nodes**, b = number of **branches**. Here $n = 3$, $b = 4$.

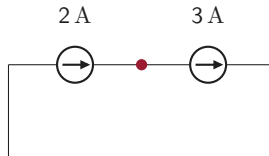


Why is one KCL redundant? Each branch current leaves one node and enters another, so it appears in exactly two node equations — once with $+$, once with $-$. Add all n of them and every current cancels: $0 = 0$. Whichever node you drop, its equation can be rebuilt from the rest, so only $n - 1$ say anything new.

Some diagrams mean nothing at all



KVL: $5 = 9$

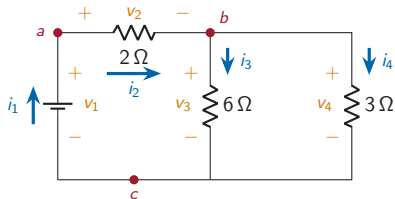


KCL: $2 = 3$

Both are **false whatever the other variables do** — the diagram describes no circuit. An ideal source promises to hold its value at any cost, and two such promises can disagree.

Give each source its internal resistance and the equations agree again, with a very large current. Sometimes that is harmless; sometimes it is why one of them makes smoke.

All 2b equations, once

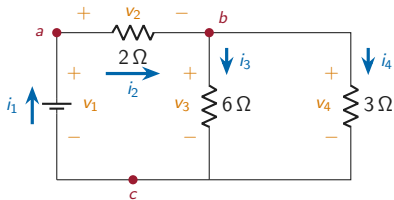


terminal (4)

$$v_1 = 12 \quad v_3 = 6 i_3$$

$$v_2 = 2 i_2 \quad v_4 = 3 i_4$$

All 2b equations, once



terminal (4)

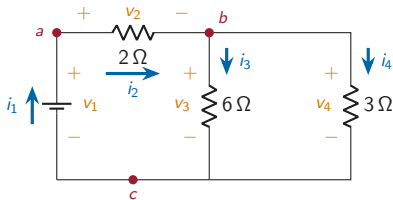
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KCL (2)

$$a: -i_1 + i_2 = 0 \quad b: -i_2 + i_3 + i_4 = 0$$

All 2b equations, once



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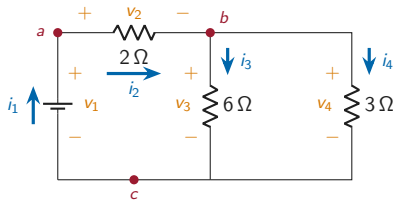
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KVL (2)

$$L_1: -v_1 + v_2 + v_3 = 0$$

$$L_2: -v_3 + v_4 = 0$$

$$i_1 = i_2 = 3 \text{ A}, \quad i_3 = 1 \text{ A}, \quad i_4 = 2 \text{ A}$$

$$v_2 = v_3 = v_4 = 6 \text{ V}$$

Free check: power must balance

$$\sum_k p_k(t) = 0 \quad \text{in any circuit, at any instant}$$



Costs one line and catches most sign errors.

Summary

KCL	$\sum \pm i_k = 0$	at every node, every instant
KVL	$\sum \pm v_k = 0$	round every loop, every instant

- $2b$ unknowns = b terminal + $(n - 1)$ KCL + $(b - n + 1)$ KVL.
- KCL holds on any closed surface, not only at a node.
- KVL says voltage forgets the path.
- Signs come from the reference arrows and polarities you drew. You may pick them by guessing the true directions, but you need not — a wrong guess costs nothing but a minus sign in the answer.
- Neither law knows what the elements are — both survive nonlinear and time-varying circuits untouched.

Next lecture

What a two-terminal element can be

Independent voltage and current sources, the general definition of a resistor, and the five words for describing one — all read off a curve in the v – i plane.

Questions?

Notes for this lecture: see the course repository.